CS 3244

Machine Learning

Week 5 – Bias and Variance & Overfitting

5 pre

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Recap and Forecast

Linear Regression

- Real valued output

$$J(\theta) = \frac{1}{m}\sum_{j=1}^{m}\left(\theta^\top x^{(j)} - y^{(j)}\right)^2$$

Logistic Regression

- Probability of a binary event

$$J(\theta) = \frac{1}{m}\sum_{j=1}^{m} ln(1 + e^{-y^{(j)}\theta^\top x^{(j)}})$$

Gradient Descent

- Take steps in opposite the gradient of the cost function.
- SGD: the average pointwise direction is equivalent to the gradient $-\nabla J$

Learning Outcomes for this week

- Understand the bias–variance tradeoff
- Understand why overfitting occurs and the role of the model complexity and sampled dataset in causing it
- Understand the effect of noisy targets as contributing another source of noise
- Be clear about the role of data resources towards model selection

Probability Recap

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Random Variable

Intuition

- Take on some random values
- Associate a distribution to the possible values

Example

Let X be the outcome of a die roll.

Possible values?

Possible distribution?

Expected Value

Intuition

Average over all possible outcome of X

$$\mathbb{E}[X] = \sum_{x \in X} x \Pr[X = x]$$

Example

Let X be the outcome of a fair dice roll. Then the expected value is $\mathbb{E}[X] = \frac{1}{6}(1 + 2 + 3 + 4 + 5 + 6) = 3.5$.

Variance

Intuition

Quantify how much deviation from expected value.

$$\mathrm{Var}[X] = \mathbb{E}[(X - \mathbb{E}[X])^2]$$

Example

Let X be the outcome of a fair dice roll. Then the variance is $\mathrm{Var}[X] = \frac{1}{6}\left((1-3.5)^2 + \cdots + (6-3.5)^2\right) = \frac{35}{12}$.

Point Estimate

Intuition

- Single best guess of some quantity of interest
- Dependent on data

Point Estimate

Example

Suppose we want to know the mean income of a population However it is unlikely that we can collect data from everyone in the population. Instead, we ask a subset of the population and use the sample mean.

What is the quantity of interest?

What is the single best guess?

Point Estimate

Example

Suppose we want to predict the housing price Y from the size of the house X. Further suppose that the relationship is govern by

$$Y = \theta_0 + \theta_1 X.$$

Here, θ_0 and θ_1 are the quantities of interest. After we collect some data, we can use linear regression to generate a point estimate for θ_0 and θ_1.

Point Estimate

Warning

- Point estimate depends on data collected
- Data collected is random
- Therefore point estimate is random variable
- Make sense to talk about the expected value and variance of point estimate

Bias of Point Estimate

Intuition

The bias measures, on average, how much does the point estimate differs from the true value of the parameter.

Let $\hat{\theta}$ be the point estimate for θ. Then we define

$$\mathrm{Bias}\left[\hat{\theta}\right] = \mathbb{E}\left[\hat{\theta}\right] - \theta$$

Bias of Point Estimate

Example

Suppose we want to estimate the population mean μ. We do so by taking the average of m data $X^{(1)}, \ldots, X^{(m)}$. Thus, our point estimate for μ is

$$\hat{\mu} = \frac{1}{m} \sum_{i=1}^{m} X^{(i)}$$

Bias of Point Estimate

Example

The bias of this point estimate is $\mathrm{Bias}[\hat{\mu}] = \mathbb{E}[\hat{\mu}] - \mu$

$$\mathbb{E}[\hat{\mu}] - \mu = \left(\frac{1}{m}\sum_{i=1}^{m}\mathbb{E}[X^{(i)}]\right) - \mu$$

Since $\mathbb{E}\left[X^{(i)}\right] = \mu$, the summation term is μ and the bias is 0.

Variance of Point Estimate

Intuition

Point estimate varies according to your data. The variance of the point estimate $\mathrm{Var}[\hat{\theta}]$ quantifies that.

Variance of Point Estimate

Example

Let us try to find the variance of the point estimate $\hat{\mu}$ from the previous example. We shall assume that the population variance is σ^2. Then

$$\mathrm{Var}[\hat{\mu}] = \mathrm{Var}\left(\frac{1}{m}\sum_{i=1}^{m} X^{(i)}\right) = \frac{1}{\mathrm{m}^2}\left(\sum_{i=1}^{m} \mathrm{Var}\left[X^{(i)}\right]\right) = \frac{\sigma^2}{m}$$

Resources

MIT 18.05 - Introduction to Probability and Statistics

https://ocw.mit.edu/courses/mathematics/18-05-introduction-to-probability-and-statistics-spring-2014/

All of Statistics (Larry Wasserman)

Chapter 2 to Chapter 6 for probability recap

Chapter 7 for point estimation

The Learning Problem

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Terminologies

Target Function

True underlying function we hope to estimate.

Hypothesis Space

The set of functions that we can pick from as an estimate to the target function.

Learning Algorithm

The algorithm that decides which function from the hypothesis space to use.

Recap – Terminology

$\mathcal{H}$ The hypothesis space of a particular learner A.

f The target function, which might not necessarily be in $\mathcal{H}$.

h The function selected by the learning algorithm A. Note that $h \in \mathcal{H}$. Also denoted $\hat{f}$, where the hat denotes approximation.

$\mathcal{D}$ The set of observed data.

A The learning algorithm that takes in $\mathcal{D}$ and outputs h.

$\mathcal{A}$ The set of all learning algorithms. We pick $A \in \mathcal{A}$. Each A has an inductive learning bias, otherwise we cannot learn (generalize) beyond $\mathcal{D}$. An

Terminology

Example

Suppose we collected the data

$$\mathcal{D} = \{(0,0), (0.04, 0.04), (0.12, 0.12)\}$$

which are generated by some unknown target function f. Let our hypothesis space $\mathcal{H} = \{h_1, h_2\}$, with $h_1(x) = \sin x$ and $h_2(x) = \cos x$. Then, our learning algorithm $\mathcal{A}$ must pick some $h \in \mathcal{H}$ that it thinks best describes $\mathcal{D}$.

An example of A is the use of linear regression with gradient descent to minimize the mean square error.

Caveats

Warning 1

$\mathcal{H}$ might not include the target function f

Warning 2

h is a random variable, dependent on the set of data $\mathcal{D}$ it gets to learn from (also assumed to be randomly sampled)

Bias and Variance, Illustrated

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Example: univariate sine target

$f: [-1,1] \to \mathbb{R}$ $\qquad$ $f(x) = \sin(\pi x)$

Only two training examples! $m = 2$

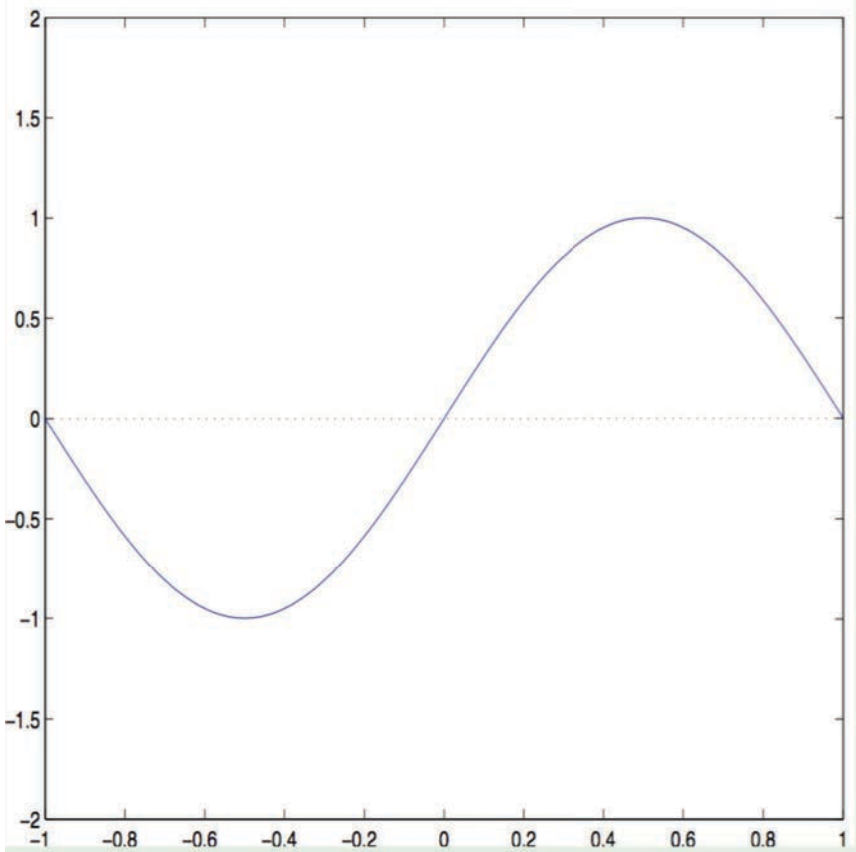
Two models used for learning:

$\mathcal{H}_0$: $h(x) = \theta_0$

$\mathcal{H}_1$: $h(x) = \theta_1 x + \theta_0$

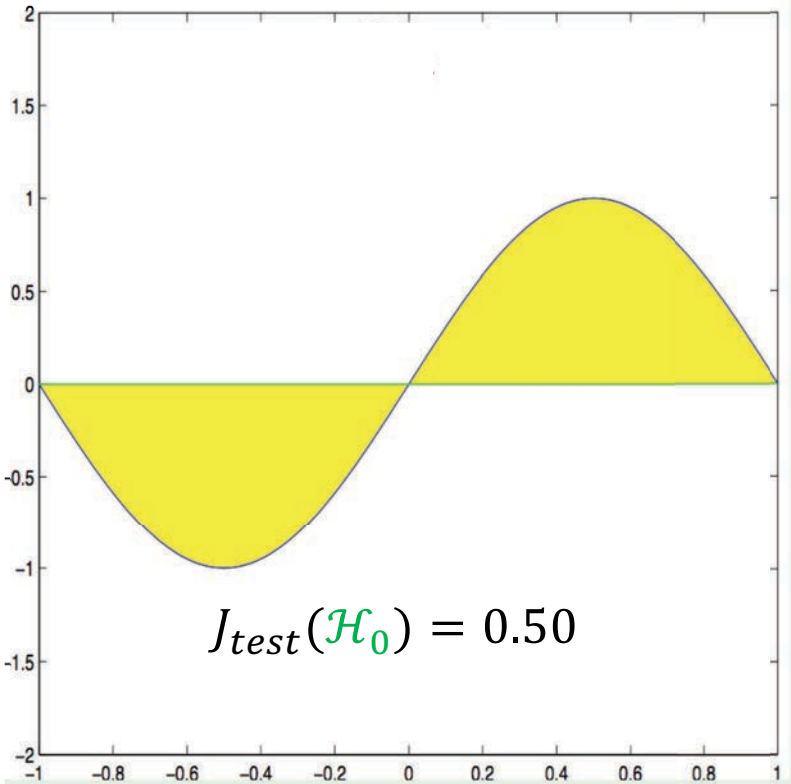
Which is better, $\mathcal{H}_0$ or $\mathcal{H}_1$?

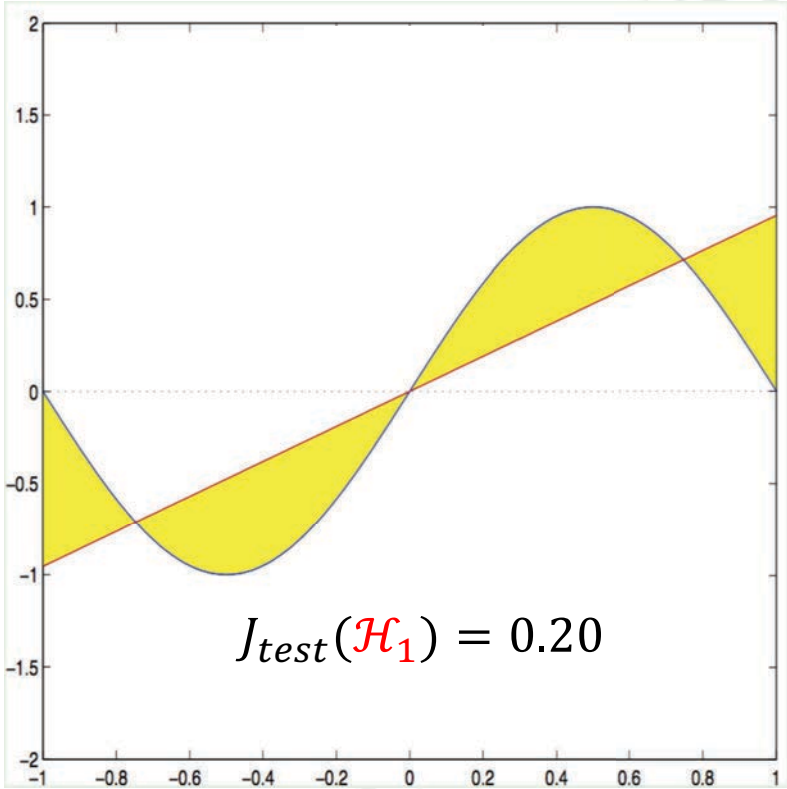
Better for <u>what</u>?



Approximation – $\mathcal{H}_0$ versus $\mathcal{H}_1$

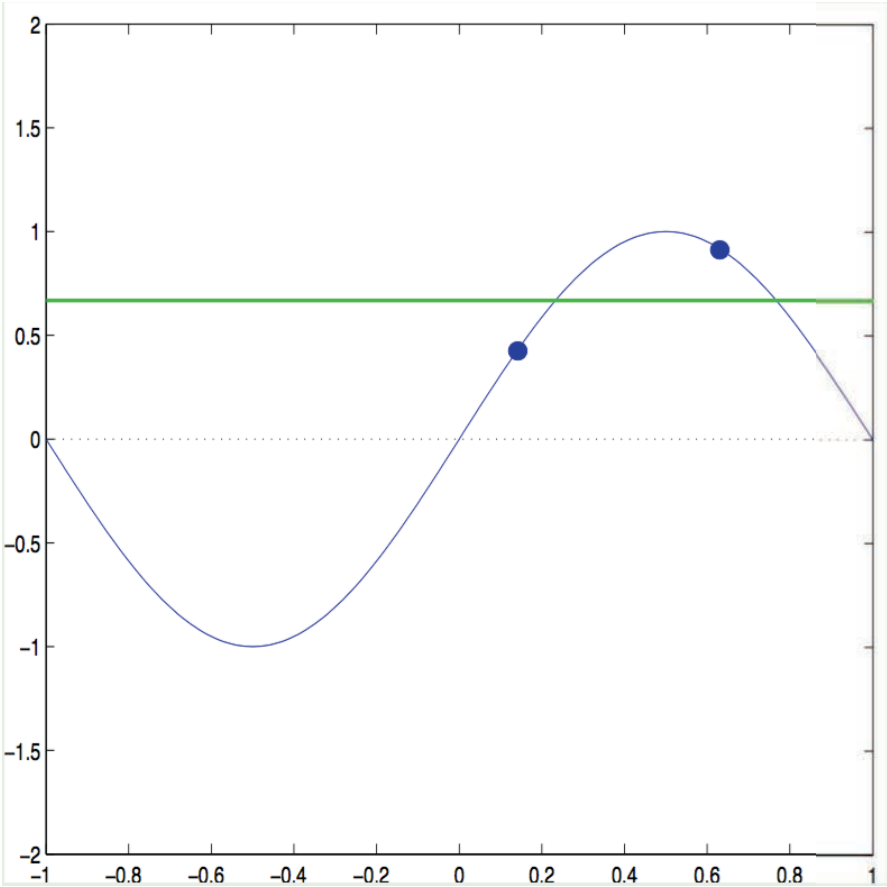
Use the full power of the model

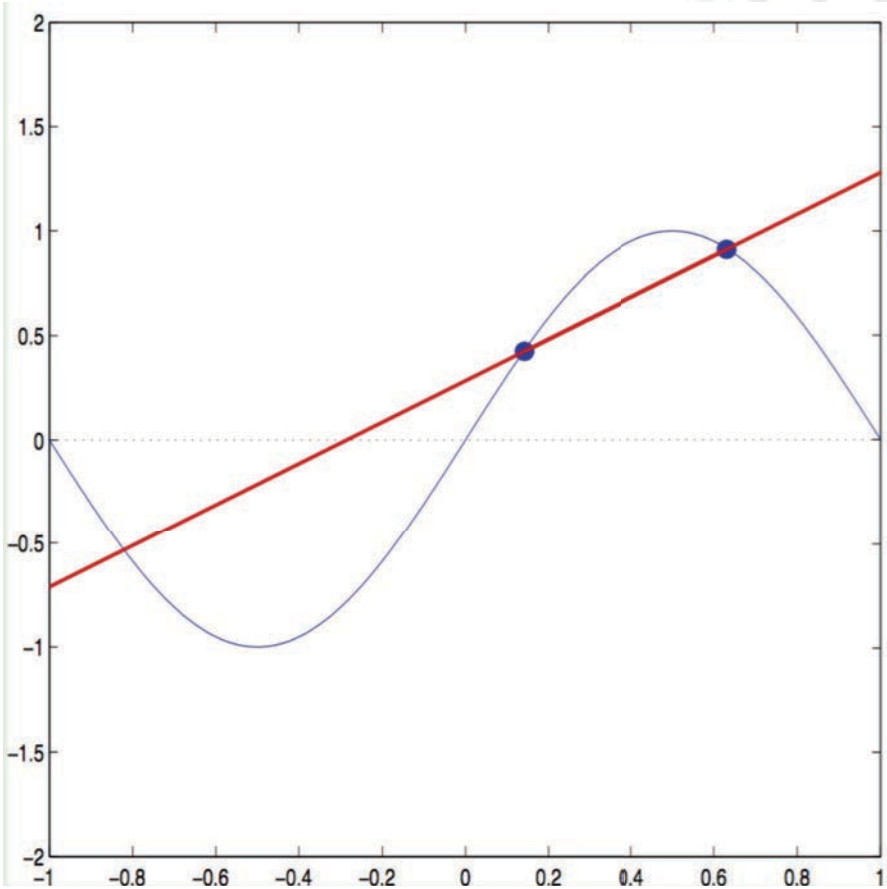




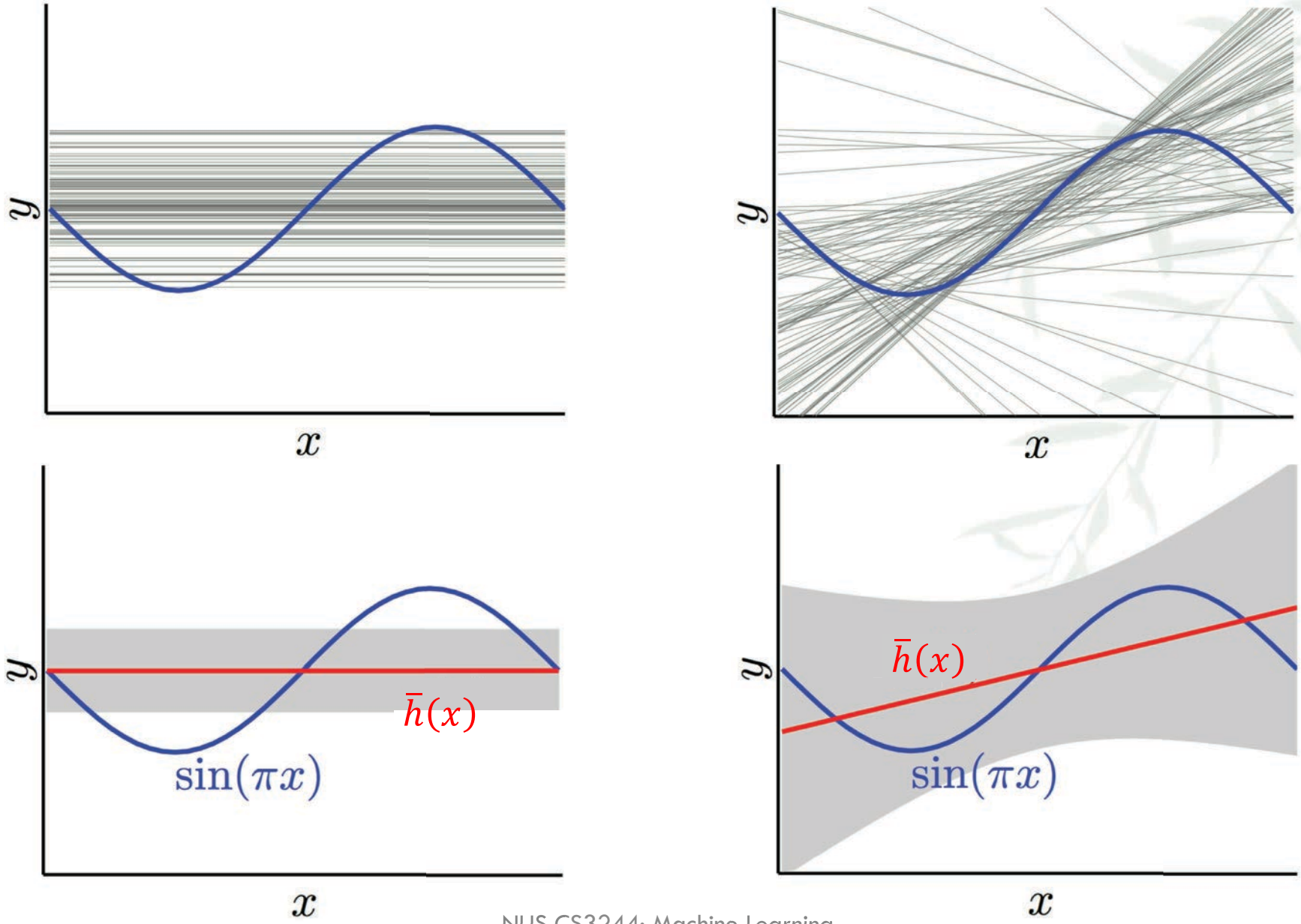
where — and — are the $\bar{h}(x)$ of each $\mathcal{H}$.

Learning – $\mathcal{H}_0$ versus $\mathcal{H}_1$



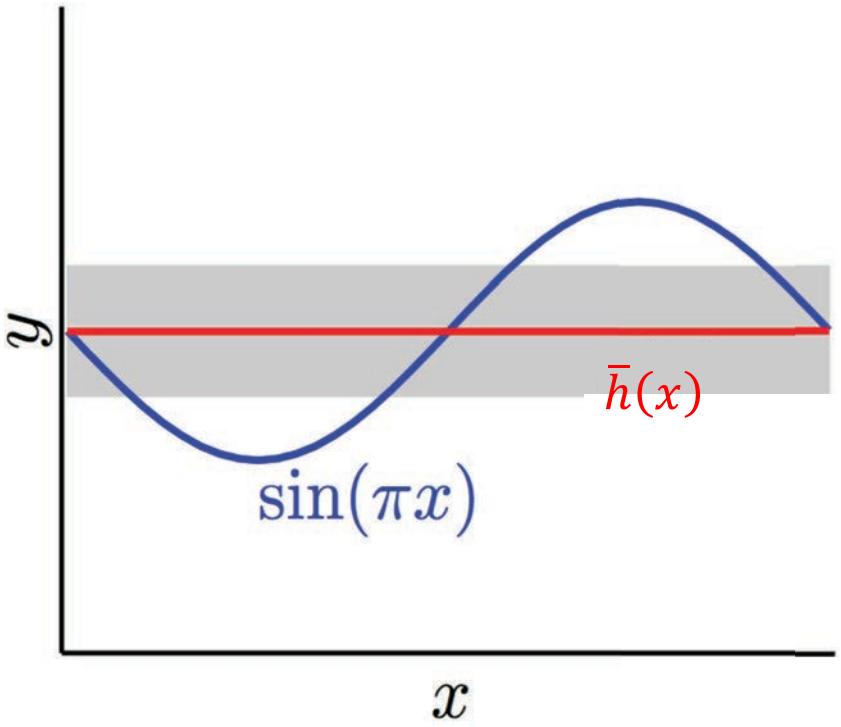


Bias & variance - $\mathcal{H}_0$ versus $\mathcal{H}_1$



And the winner is …

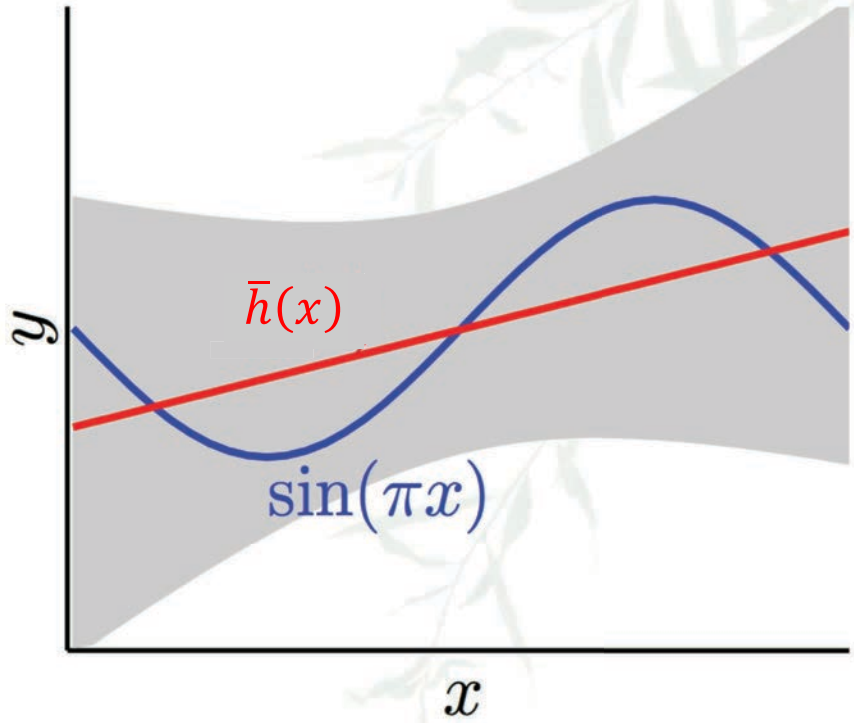
$\mathcal{H}_0$



Bias = 0.50

Var = 0.25

$\mathcal{H}_1$



Bias = 0.21

Var = 1.69

Lesson learned

Match the 'model complexity' to the

data resources, not to the **target complexity**.

Quality and Quantity

Over fitting

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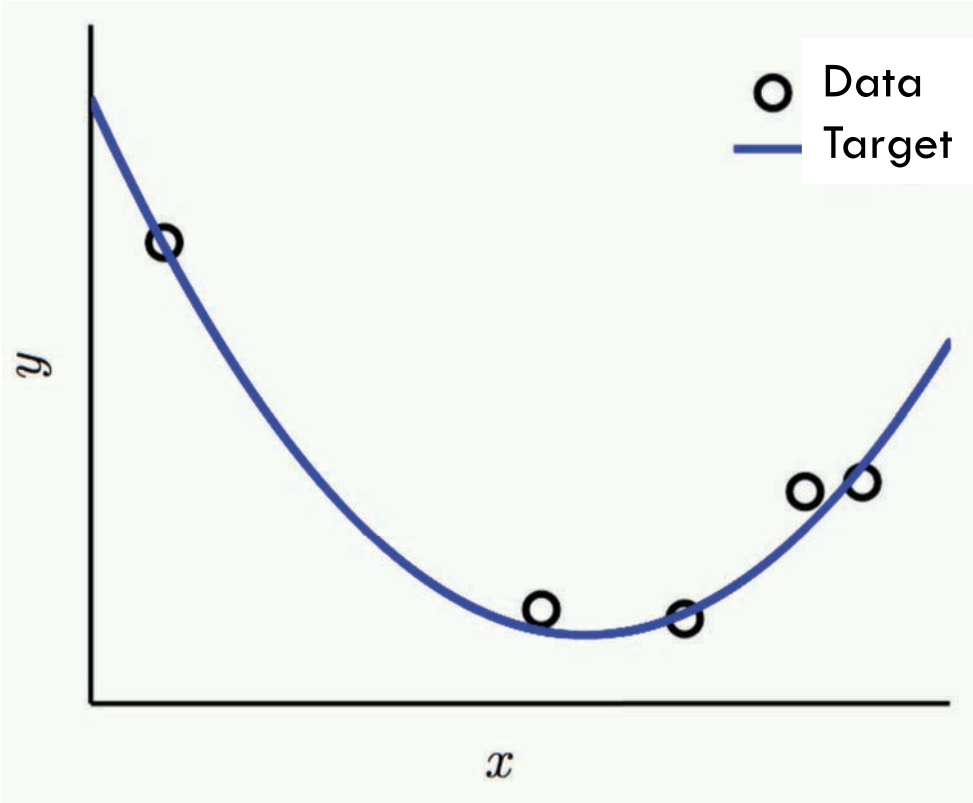
Overfitting, Illustrated

Simple target function

5 data points –
slightly noisy

4th-order polynomial fit

$J_{train} = 0$ but J_{test} is huge



Overfitting, Illustrated

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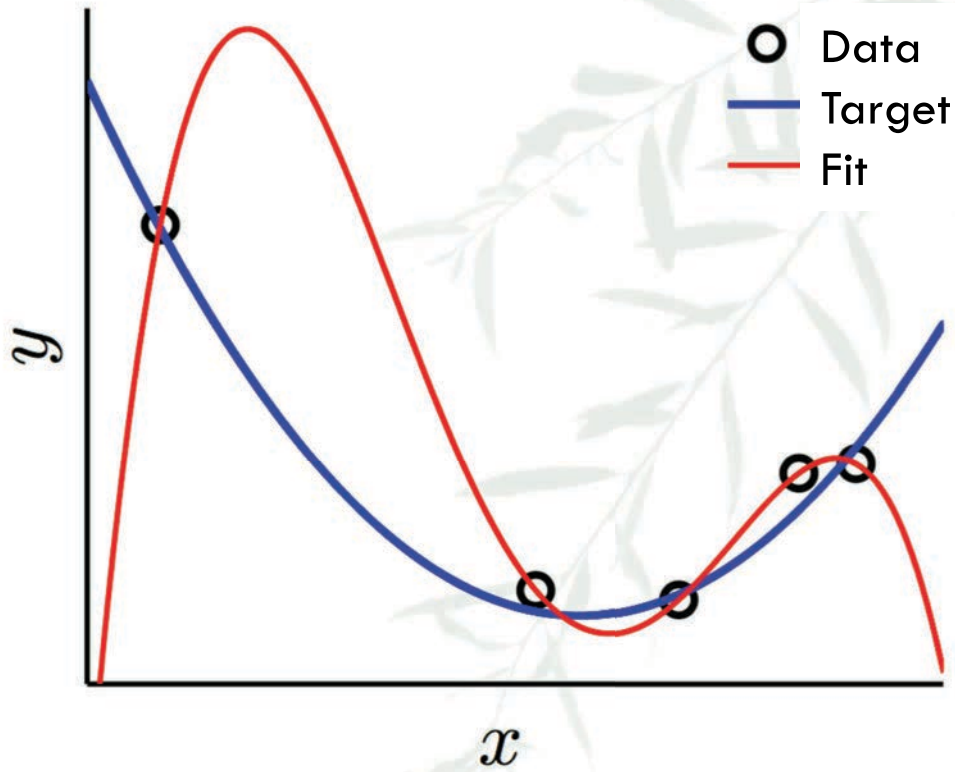


The culprit

Overfitting:

"Fitting the data more than is warranted"

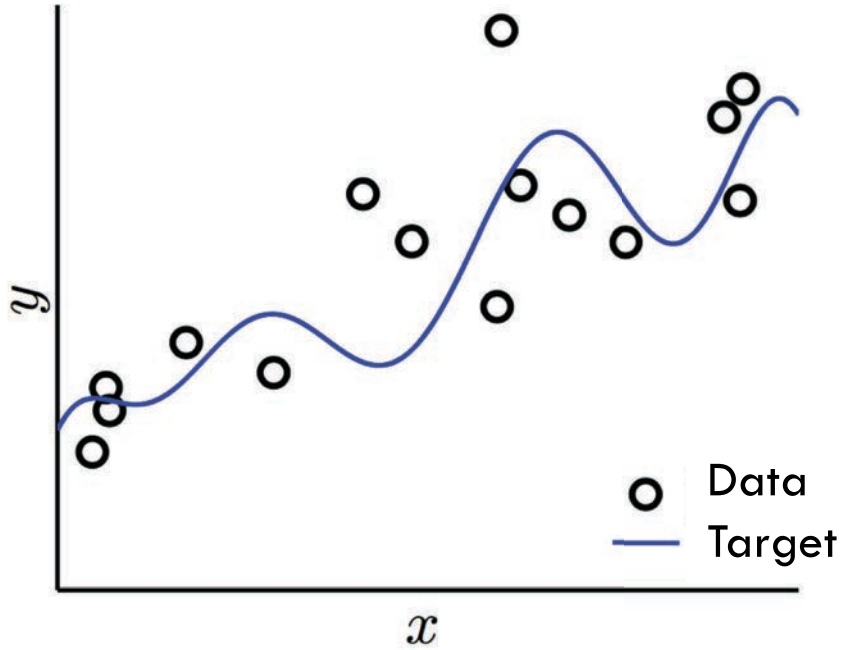
Culprit: fitting the noise.

Not just useless,

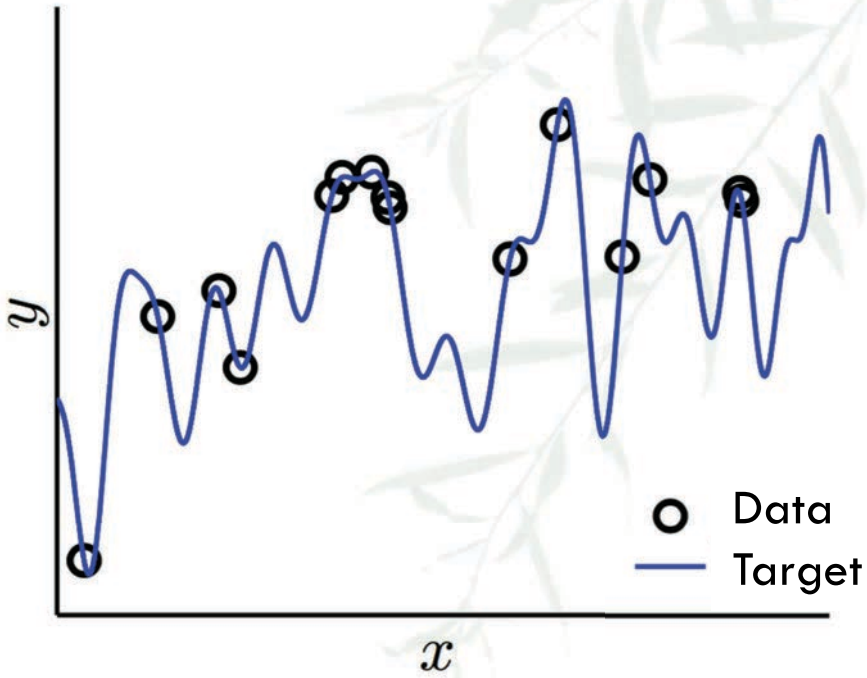
but harmful

Case study: 2nd versus 10th order polynomials

10th-order target + noise



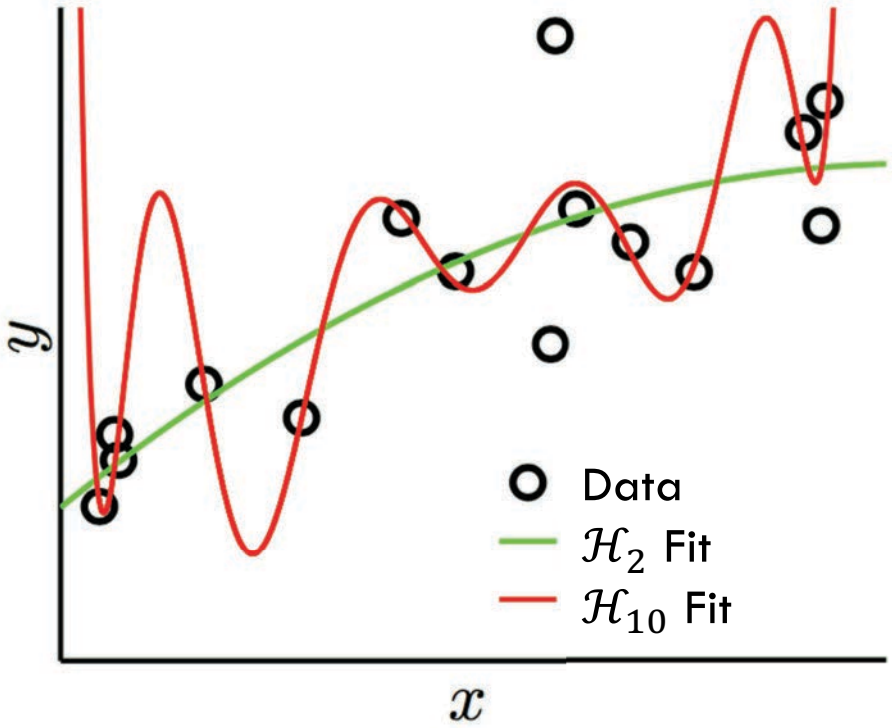
50th order target, noiseless



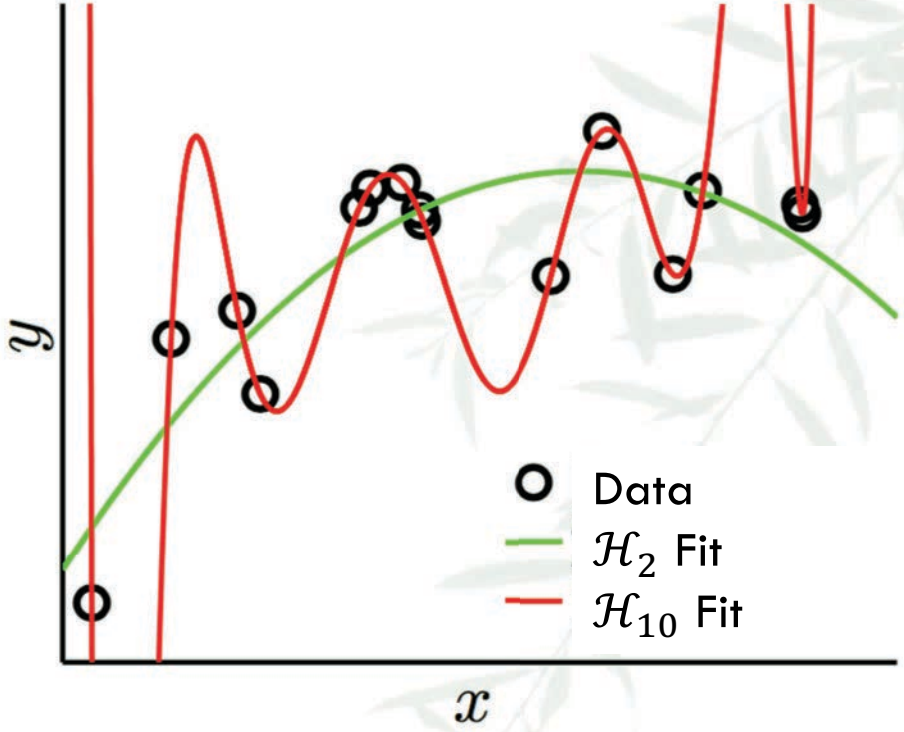
Given: 15 data points

Poll: pick your model of choice for each scenario

Two fits for each function



Noisy low-order target



Noiseless high-order target

	2nd Order	
J_{train}	0.050	
J_{test}	0.127	

	2nd Order	
J_{train}	0.029	
J_{test}	0.120	

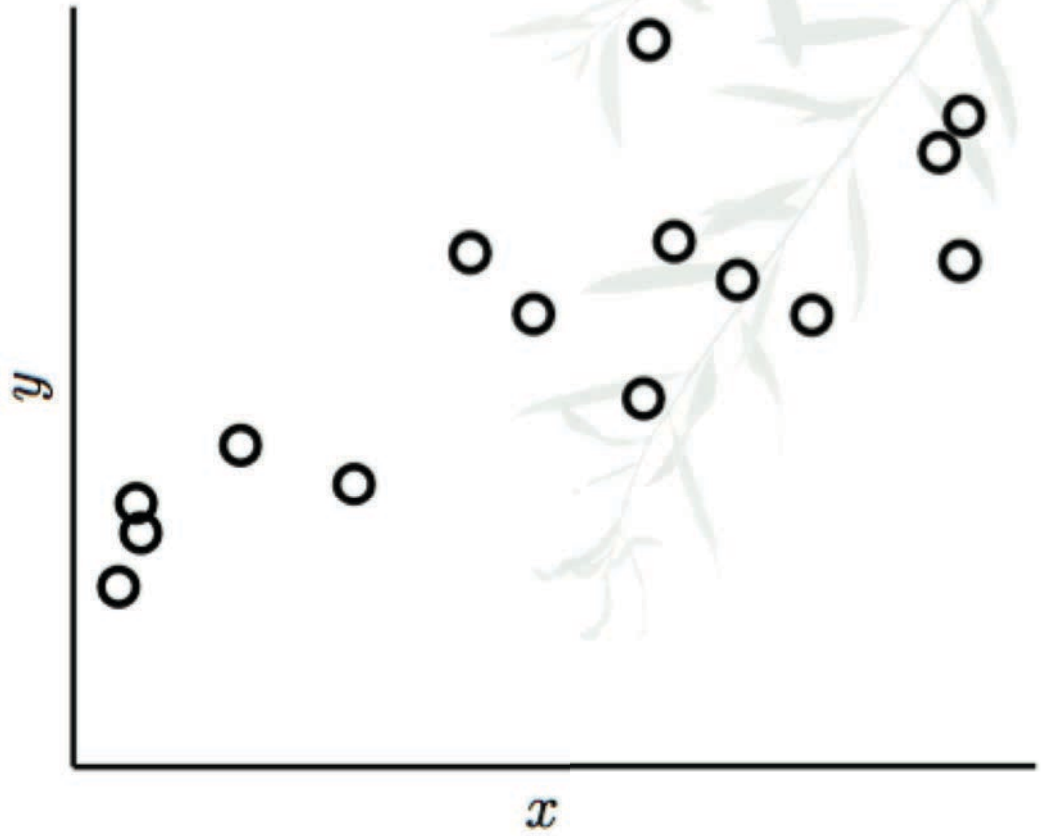
Two learners

Two learners Overfitting and Restrained:

Both <u>know</u> the target is 10th order,

you get 15 data points.

O chooses $\mathcal{H}_{10}$

R chooses $\mathcal{H}_2$

Who wins?